Research Notes on Linear regression

Linear regression

$$\hat{y} = X\beta + \varepsilon$$

$$\hat{\beta} = (X^T X)^{-1} X^T y$$

>> Section 1

Finance The capital asset pricing model uses linear regression as well as the concept of beta for analyzing and quantifying the systematic risk of an investment. This comes directly from the beta coefficient of the linear regression model that relates the return on the investment to the return on all risky assets.

>> Section 2

The simplest case of a single scalar predictor variable x and a single scalar response variable y is known as simple linear regression. The extension to multiple and/or vector-valued predictor variables (denoted with a capital X) is known as multiple linear regression, also known as multivariable linear regression (not to be confused with multivariate linear regression). Multiple linear regression is a generalization of simple linear regression to the case of more than one independent variable, and a special case of general linear models, restricted to one dependent variable. The basic model for multiple linear regression is

>> Section 3

Parameters β j {\displaystyle \beta _{j}} in the original model, including β 0 {\displaystyle \beta _{0}} , are simple functions of β j ′ {\displaystyle \beta _{j}'} in the standardized model. The standardization of variables does not change their correlations, so { x 1 ′ , x 2 ′ , … , x q ′ } {\displaystyle \{x_{1}',x_{2}',\dots ,x_{q}'\}} is a group of strongly correlated variables in an APC arrangement and they are not strongly correlated with other predictor variables in the standardized model. A group effect of { x 1 ′ , x 2 ′ , … , x q ′ } {\displaystyle \{x_{1}',x_{2}',\dots ,x_{q}'\}} is

>> Section 4

X {\displaystyle \mathbf {X} } may be seen as a matrix of row-vectors x i ⋅ {\displaystyle \mathbf {x} _{i\cdot }} or of n-dimensional column-vectors x ⋅ j {\displaystyle \mathbf {x} _{\cdot j}} , which are known as regressors, exogenous variables, explanatory variables, covariates, input variables, predictor variables, or independent variables (not to be confused with the concept of independent random variables). The matrix X {\displaystyle \mathbf {X} } is sometimes called the design matrix. Usually a constant is included as one of the regressors. In particular, x i 0 = 1 {\displaystyle x_{i0}=1} for i = 1 , … , n {\displaystyle i=1,\ldots ,n} . The corresponding element of β is called the intercept. Many statistical inference procedures for linear models require an intercept to be present, so it is often included even if theoretical considerations suggest that its value should be zero. Sometimes one of the regressors can be a non-linear function of another regressor or of the data values, as in polynomial regression and segmented regression. The model remains linear as long as it is linear in the parameter vector β. The values xij may be viewed as either observed values of random variables Xj or as fixed values chosen prior to observing the dependent variable. Both interpretations may be appropriate in different cases, and they generally lead to the same estimation procedures; however different approaches to asymptotic analysis are used in these two situations.

>> Section 5

β {\displaystyle {\boldsymbol {\beta }}} is a (p + 1) {\displaystyle (p+1)} -dimensional parameter vector, where β 0 {\displaystyle \beta _{0}} is the intercept term (if one is included in the model—otherwise β {\displaystyle {\boldsymbol {\beta }}} is p-dimensional). Its elements are known as effects or regression coefficients (although the latter term is sometimes reserved for the estimated effects). In simple linear regression, p=1, and the coefficient is known as regression slope. Statistical estimation and inference in linear regression focuses on β. The elements of this parameter vector are interpreted as the partial derivatives of the dependent variable with respect to the various independent variables.

>> Section 6

Regularized Regression Ridge regression and other forms of penalized estimation, such as Lasso regression, deliberately introduce bias into the estimation of β in order to reduce the variability of the estimate. The resulting estimates generally have lower mean squared error than the OLS estimates, particularly when multicollinearity is present or when overfitting is a problem. They are generally used when the goal is to predict the value of the response variable y for values of the predictors x that have not yet been observed. These methods are not as commonly used when the goal is inference, since it is difficult to account for the bias.

>> Section 7

ξ (w) = w 1 β 1 + w 2 β 2 + ■ + w q β q , {\displaystyle \xi (\mathbf {w})=w_{1}\beta _{1}+w_{2}\beta _{2}+\dots +w_{q}\beta _{q},}

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>> Section 8

I (D , β →) = log ■ ∏ i = 1 n P r (y i | x i → ; β → , σ) = log ■ ∏ i = 1 n 1 2 π σ exp ■ (− (y i − β → · x i →) 2 2 σ 2) = n log ■ 1 2 π σ − 1 2 σ 2 ∑ i = 1 n (y i − β → · x i →) 2 {\displaystyle {\begin{aligned}I(D,{\vec {\beta }})&=\log \prod _{i=1}^{n}Pr(y_{i}|{\vec {x_{i}}}\,\,;{\vec {\beta }},\sigma)\\&=\log \prod _{i=1}^{n}{\frac {1}{{\sqrt {2\pi }}\sigma }}\exp \left(-{\frac {\left(y_{i}-{\vec {\beta }}\,\cdot \,{\vec {x_{i}}}\right)^{2}}{2\sigma ^{2}}}\right)\\&=n\log {\frac {1}{{\sqrt {2\pi }}\sigma }}-{\frac {1}{2\sigma ^{2}}}\sum _{i=1}^{n}\left(y_{i}-{\vec {\beta }}\,\cdot \,{\vec {x_{i}}}\right)^{2}\end{aligned}}}

>> Section 9

y i ≈ ∑ j = 0 m β j × x j i = β → · x i → {\displaystyle y_{i}\approx \sum _{j=0}^{m}\beta _{j}\times x_{j}^{i}={\vec {\beta }}\cdot {\vec {x_{i}}}} . In the least-squares setting, the optimum parameter vector is defined as such that minimizes the sum of mean squared loss:

>> Section 10

y i ≈ β 0 + ∑ j = 1 m β j × x j i {\displaystyle y_{i}\approx \beta _{0}+\sum _{j=1}^{m}\beta _{j}\times x_{j}^{i}} . If x i → {\displaystyle {\vec {x_{i}}}} is extended to x i → = [1 , x 1 i , x 2 i , … , x m i] {\displaystyle {\vec {x_{i}}}=\left[1,x_{1}^{i},x_{2}^{i},\ldots ,x_{m}^{i}\right]} then y i {\displaystyle y_{i}} would become a dot product of the parameter and the independent vectors, i.e.

>> Section 11

Y i = β 0 + β 1 X i 1 + β 2 X i 2 + … + β p X i p + ■ i {\displaystyle Y_{i}=\beta _{0}+\beta _{1}X_{i1}+\beta _{2}X_{i2}+\ldots +\beta _{p}X_{ip}+\epsilon _{i}}